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REM sleep reduction leads to fear overgeneralization via negatively modulating prefrontal theta oscillations

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Abstract

Background Fear overgeneralization, defined as the excessive fear response to non-threatening stimuli, is a hallmark of anxiety-related disorders. Sleep has long been recognized as a critical factor in the consolidation and processing of fear memories, with research suggesting that different sleep phases, particularly rapid eye movement (REM) and non-rapid eye movement (NREM) sleep, may have distinct roles. However, how sleep influences fear generalization remains largely unknown. We systematically investigated the dynamic effects of sleep phases and their associated neural oscillations on fear generalization recently after sleep manipulation and remotely 1 week later.

Methods In a randomized controlled between-subjects experiment, 126 participants were assigned to the four groups: total sleep deprivation (SD), early-night (dominated by NREM sleep) SD, late-night (dominated by REM sleep) SD, and total sleep. Participants completed the fear conditioning test before the sleep-manipulated night, the recent fear generalization test after sleep, and the remote fear generalization test 1 week later. Both neuroimaging data and behavioral data (subjective risk ratings and objective skin conductance responses) were collected synchronously to analyze brain functional changes during generalized stimuli processing.

Results Sleep, particularly REM sleep phase, inhibits fear generalization, whereas late-night SD, by reducing REM sleep, induces fear overgeneralization comparable to total SD. The percentage of REM sleep was negatively associated with the degree of fear generalization and positively correlated with dorsolateral prefrontal cortex BOLD activity during the recent generalization test. Notably, prefrontal theta oscillations mediated 44% of REM sleep's effects on fear generalization.

Conclusions These findings demonstrate the essential role of post-conditioning REM sleep, particularly its reliance on theta oscillations in the frontal lobe, in mitigating fear overgeneralization.

Keywords Sleep, REM sleep, Fear generalization, Dorsolateral prefrontal cortex, Theta oscillations

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Background

Fear represents a fundamental adaptive mechanism that enables effective generalization of learned threat cues and react to new but similar situations [1]. However, this evolutionarily conserved process becomes maladaptive when fear responses excessively extend to safe stimuli resembling actual threat cues, contributing to the development of anxiety and stress-related disorders such as post-traumatic stress disorder (PTSD) and specific phobias [2, 3]. Research reveal that overgeneralization-prone individuals exhibit persistent distress when encountering trauma-associated cues, even in non-threatening contexts [4]. Crucially, among various modulators of fear generalization, sleep emerges with unique relevance due to its bidirectional regulatory role in both adaptive generalization processes and maladaptive overgeneralization patterns. This dual role manifests both temporally, where post-trauma sleep disruption predicts subsequent psychopathology development [5], and symptomatically, sleep problems are a core feature of those symptoms and can affect the onset, maintenance, and recovery [6]. Advancing our understanding of how sleep influencs fear generalization and its underlying mechanisms will facilitate the development of targeted intervention strategies for symptoms associated with overgeneralization.

Sleep is increasingly recognized the role of fear processing and regulation [7–9]. Supporting this, prior research has shown that individuals deprived of sleep before fear acquisition exhibit heightened fear responses [10]. Notably, the rapid eye movement (REM) sleep, which predominantly occurs during the late half of the night, has been implicated in the consolidation of emotional memories [11, 12]. For instance, the amount of REM sleep following fear conditioning correlates with both fear recall and extinction [13, 14]. Importantly a negative relationship has been observed between the duration of REM sleep and the skin conductance response (SCR) to conditioned fear stimuli 24 h later. While substantial evidence underscores the role of sleep in fear learning, studies examining its effects on fear generalization are limited and yield mixed findings [15, 16]. For example, one study reported no differences in the extent of fear generalization between nap and wake groups [17], while another found greater fear generalization in the nap group compared to the wake group [18]. These conflicting results leave the adaptive or maladaptive nature of sleep's influence on fear generalization unresolved. Moreover, it remains unclear which specific sleep stage is critically involved in the generalization process, highlighting the need for further investigation.

To date, numerous animal and human studies have sought to elucidate the neural basis of fear generalization using multimodal approaches. Nevertheless, critical gaps persist regarding how sleep modulates those neural activities to shape generalization processes. Neuroimaging meta-analyses consistently implicate a distributed network encompassing the prefrontal cortex, insula, hippocampus, amygdala, angular gyrus, locus coeruleus, and thalamus in fear generalization gradients—neural response patterns that mirror behavioral fear generalization slopes [19–21]. Notably, the dorsomedial prefrontal cortex and bilateral anterior insula have been correlated with fear generalization gradients [19]. Additionally, individuals with PTSD exhibit abnormal responses to fear generalization gradients in the dorsolateral prefrontal cortex (DLPFC), insula, and hippocampus [20]. Given the prefrontal cortex's critical role in processing emotional memories during sleep [22], we hypothesize that it may also be actively involved in mediating fear generalization during sleep. Further research is needed to explore this potential connection and enhance our understanding of the neural mechanism of sleep's influence on fear generalization.

Sleep-related neural oscillations, including delta, theta, alpha, and beta rhythms, orchestrate distinct neurophysiological mechanisms during sleep-phase transitions [23, 24]. Specifically, post-learning theta oscillations (4–8 Hz) are closely linked to the retrieval of fear memories and the successful extinction of fear-related memories [25–27]. The synchronization of the 4-Hz theta rhythms is highly coordinated among the limbic cortex, the prefrontal cortex, and the amygdala following fear learning, during both wakefulness and sleep [23, 25, 28]. Furthermore, theta oscillations during REM sleep have been positively correlated with the retention of conditioned fear [26, 29]. This bidirectional theta power functionality—enhancing both consolidation precision and extinction efficacy (via REM-phase plasticity)—positions theta rhythms as key oscillatory mediators regulating the adaptive-maladaptive fear generalization continuum during REM sleep. Mechanistic investigation of theta frequency-specific modulations may reveal novel targets for REM sleep-based interventions in pathological generalization.

Building on theoretical frameworks positing sleep's fear-regulatory function through memory reprocessing [30–32], we proposed two principal hypotheses: (1) sleep, especially the REM sleep stage, plays a critical role in reducing fear overgeneralization, and (2) the prefrontal cortex and theta activities would be key mediators in this procession. To test these hypotheses, we implemented a multi-timepoint fear generalization paradigm combining behavioral, neuroimaging, and polysomnographic measures. Participants underwent fear conditioning followed by generalization testing at 24-h and 7-day intervals—timepoints capturing recent generalization and remote generalization, respectively. During fear generalization

tasks, we concurrently acquired functional magnetic resonance imaging (fMRI) data and SCR to quantify neural and behavioral generalization index. Full-night polysomnography with electroencephalogram (EEG) was administered post-conditioning to characterize sleep architecture and oscillatory dynamics.

Methods

Participants

All participants ($n = 145$, 71 females and 74 males) were right-handed, healthy volunteers (age: 24.32 ± 2.91) with consistent sleep patterns, as confirmed by a 7-day sleep diary and actigraphy. They had an average sleep duration of 7–8 h per day and no history of psychiatric or neurological disorders, illegal drug use, or night shift work. The use of caffeine, alcohol, or sleep-affecting medications was prohibited within 48 h prior to the experiment. Additionally, all participants had normal or corrected-to-normal vision, and all female participants reported regular menstrual cycles and were not using oral contraceptives at the time of the study.

Fourteen participants were excluded due to insufficient fear memory acquisition (SCR elicited by the final unconditioned CS+ versus CS− less than 0.05 μS [33]). Five participants were excluded for showing a uniform response throughout or missing more than 20% of the stimuli. Two participants were excluded due to head movements exceeding three voxels in amplitude during the fMRI task. The final analysis included 126 participants (61 females and 65 males).

Split-night sleep procedure

Participants were randomly assigned to one of four groups for sleep manipulation (see Table 1), with two groups following a split-night paradigm [34] that involved early-night and late-night sleep deprivation groups. Both groups were allocated a total sleep opportunity of 4.5 h [12]. Participants in the early-night sleep deprivation group slept from 03:30 to 07:30 and remained awake from 23:00 to 03:30. This schedule was designed to ensure that participants experienced predominantly NREM sleep in the first half of the night and stayed awake during the second half. Participants in the late-night sleep deprivation group stayed awake from 03:00 to 07:30 and slept from 23:00 to 03:00. This schedule was intended to ensure that participants experienced predominantly REM sleep during the latter half of the night, remaining awake in the first half. While awake, participants were allowed to engage in quiet activities such as reading printed books or paper materials and walking indoors. They were also permitted to drink water in moderation. These activities were chosen to maintain a low level of physical and cognitive stimulation.

Table 1 Demographic characteristics

	Sleep deprivation group ($n = 16$)	Total sleep group ($n = 36$)	Early-night sleep group ($n = 38$)	Late-night sleep group ($n = 36$)	*p* value
Age (years)	22.00 ± 2.25	23.53 ± 2.21	22.84 ± 2.59	23.00 ± 2.42	0.20
Female (*n*, %)	9 (56%)	21 (58%)	16 (42%)	15 (42%)	0.71
BMI	22.33 ± 3.34	22.01 ± 3.05	22.29 ± 3.03	21.66 ± 2.28	0.78
PSQI score	2.94 ± 1.29	3.50 ± 1.48	3.03 ± 1.55	3.36 ± 1.50	0.43
MEQ score	49.88 ± 6.58	49.72 ± 5.69	51.32 ± 5.75	51.33 ± 5.60	0.53
STAI-T score	36.00 (± 8.22)	35.81 (± 8.72)	33.00 (± 5.80)	35.62 (± 8.47)	0.41
STAI-S score	33.06 (± 10.81)	31.36 (± 8.22)	28.59 (± 6.01)	32.47 (± 9.47)	0.19
BDI score	26.06 (± 4.77)	24.62 (± 4.89)	23.58 (± 3.22)	24.67 (± 5.46)	0.34
GHQ-12 score	21.19 (± 4.05)	21.89 (± 5.28)	20.97 (± 6.88)	20.56 (± 5.83)	0.81
SAS score	30.44 (± 5.59)	29.14 (± 3.56)	28.03 (± 3.89)	28.97 (± 4.17)	0.26
PCL-5 neuroticism	23.38 (± 5.94)	20.38 (± 6.39)	18.92 (± 6.64)	20.08 (± 7.02)	0.17
PCL-5 conscientiousness	33.75 (± 5.41)	36.59 (± 5.74)	38.13 (± 5.29)	36.33 (± 5.44)	0.07
PCL-5 agreeableness	35.81 (± 4.25)	36.57 (± 5.00)	37.63 (± 4.52)	36.19 (± 5.47)	0.52
PCL-5 openness	33.25 (± 5.77)	35.92 (± 5.93)	36.47 (± 5.11)	35.44 (± 5.45)	0.27
PCL-5 extraversion	29.31 (± 6.26)	31.22 (± 7.11)	30.39 (± 5.93)	32.03 (± 6.01)	0.49

Values are presented as mean ± SD

BMI body mass index, *PSQI* Pittsburgh Sleep Quality Index, *MEQ* Horne-Ostberg Morningness-Eveningness Questionnaire, *STAI-T* State-Trait Anxiety Inventory-Trait, *STAI-S* State-Trait Anxiety Inventory-State, *BDI* Beck Depression Inventory, *GHQ-12* General Health Questionnaire-12 items, *SAS* Self-Rating Anxiety Scale, *PCL-5* Posttraumatic Stress Disorder Checklist for DSM-5

Experimental protocol

After completing questionnaire screenings and 5 days of actigraphy monitoring (Actiwatch Spectrum PRO, Philips Respironics, Inc., Murrysville, PA, Oregon), participants arrived at the lab for two consecutive nights.

On night 1 (habituation), participants arrived at 21:00, were fitted with the PSG system, went to bed at 23:00, and were awakened and had the PSG apparatus removed at 7:30 the following morning.

On night 2 (sleep manipulation), the late-night sleep deprivation sleep group (Late-SD, $n=38$) slept during first half of the night asleep (23:00 to 3:30, 4.5 h sleep) and remained awake during the second half (3:30 to 7:30). Early-night sleep deprivation group (Early-SD, $n=36$) stayed awake during the first half of the night (23:00 to 3:00) and slept during the second half of the night (3:00 to 7:30, 4.5 h sleep). The total sleep group (TS, $n=36$) slept throughout the entire night (23:00 to 7:30, 8 h sleep). The total sleep deprivation group (SD, $n=16$) stayed awake the entire night (23:00 to 7:30, 0 h sleep) and were allowed to engage in activities such as reading magazines and extracurricular books.

On day 1 evening, participants underwent a fear acquisition session with SCR measurements between 19:00 and 21:00, while functional magnetic resonance imaging (fMRI) was conducted on participants in the three sleep groups (Early-SD, Late-SD, TS) and SCR-only for SD group.

On day 2 (recent generalization test), participants completed the recent generalization test the following morning (8:00–10:00). The Early-SD, Late-SD, and TS groups underwent fMRI scanning during this test. Only behavioral data was collected from the SD group.

On day 8 (remote generalization test), participants completed the remote generalization test in the morning (8:00–10:00) 1 week later, replicating the recent test protocol.

The experimental protocol is shown in Fig. 1A.

Fear acquisition

Participants were introduced to two visual stimuli: a small circle and a large circle. One of these stimuli was paired with an aversive electric shock (unconditioned stimulus, US), designating it as the CS+ (fear conditioned stimulus), while the other served as the CS− (safe conditioned stimulus, no shock). In the pre-acquisition phase, both CS+ and CS− were presented three times without any accompanying US. SCR were recorded to ensure there was no pre-existing bias in the participants' response to the stimuli. During the fear acquisition phase, both the CS+ and CS− stimuli were presented 12 times, with each presentation lasting 8 s. The order of presentation was pseudorandomized, with each stimulus beginning with a fixation cross (1 to 3 s), followed by the visual presentation of the CS (approximately 1000 ms) and blank intervals ranging from 10 to 16 s. The CS+ was followed by the US in 75% of presentations, whereas the CS− was never followed by a shock. Mild electrical shocks (150 ms duration) were delivered via a cable attached to the participant's left inner wrist using an STM 200 constant-current stimulator (BIOPAC Systems, Goleta, CA, USA). Prior to the experiment, each participant adjusted the intensity of the electric shock to a level that was uncomfortable but not painful. Participants were informed that each stimulus might or might not be followed by an electrical shock, and they were asked to provide button rating responses indicating how likely they thought a shock would follow each stimulus (Fig. 1B).

Fear generalization

To assess generalization, participants were shown the CS+, CS−, and a set of 8 novel circles that gradually varied in size between the two original circles (GS, generalization stimuli). Participants were expected to display a fear response not only to the CS+ but also to stimuli visually similar to it, indicating a generalized response. By measuring the SCR and self-reported fear ratings for each stimulus, we could assess how far the fear response had generalized from the original CS+ based on the size

(See figure on next page.)

Fig. 1 Experimental design. **A** Upon arriving at the laboratory, participants experienced a night of sleep adaptation (night 1) followed by a night of manipulation (night 2). The experimental timeline included a fear acquisition phase before sleep (day 1) and a fear generalization test after waking (recent test on day 2) and again 1 week later (remote test on day 8). Sleep patterns were monitored over the two consecutive nights using sleep wristbands, sleep diaries, and polysomnography. **B** During the fear acquisition phase, conditioned stimuli (CS+ and CS−) were assigned as the largest and smallest circles, respectively, with their selection counterbalanced among participants. Only CS+ and CS− were presented, requiring participants to learn fear associations between CS+ and an electric shock. **C** In the fear generalization phase, all stimuli, including the GS, were presented to participants, prompting them to respond. Throughout these phases, task-related neuroimaging data and behavioral measures (subjective fear ratings, objective skin conductance responses) were collected. Gaussian width (σ) was used as an index of fear generalization. **D** The four groups included: SD, the total sleep deprivation group; Early-SD, the early-night sleep deprivation group; Late-SD, the late-night sleep deprivation group; and TS, the total sleep group. CS+ denotes fear conditioned stimulus; CS− represents the safe conditioned stimulus; and GS stands for the generalization stimulus. PSG, polysomnography; fMRI, functional magnetic resonance imaging; SCR, skin conductance response

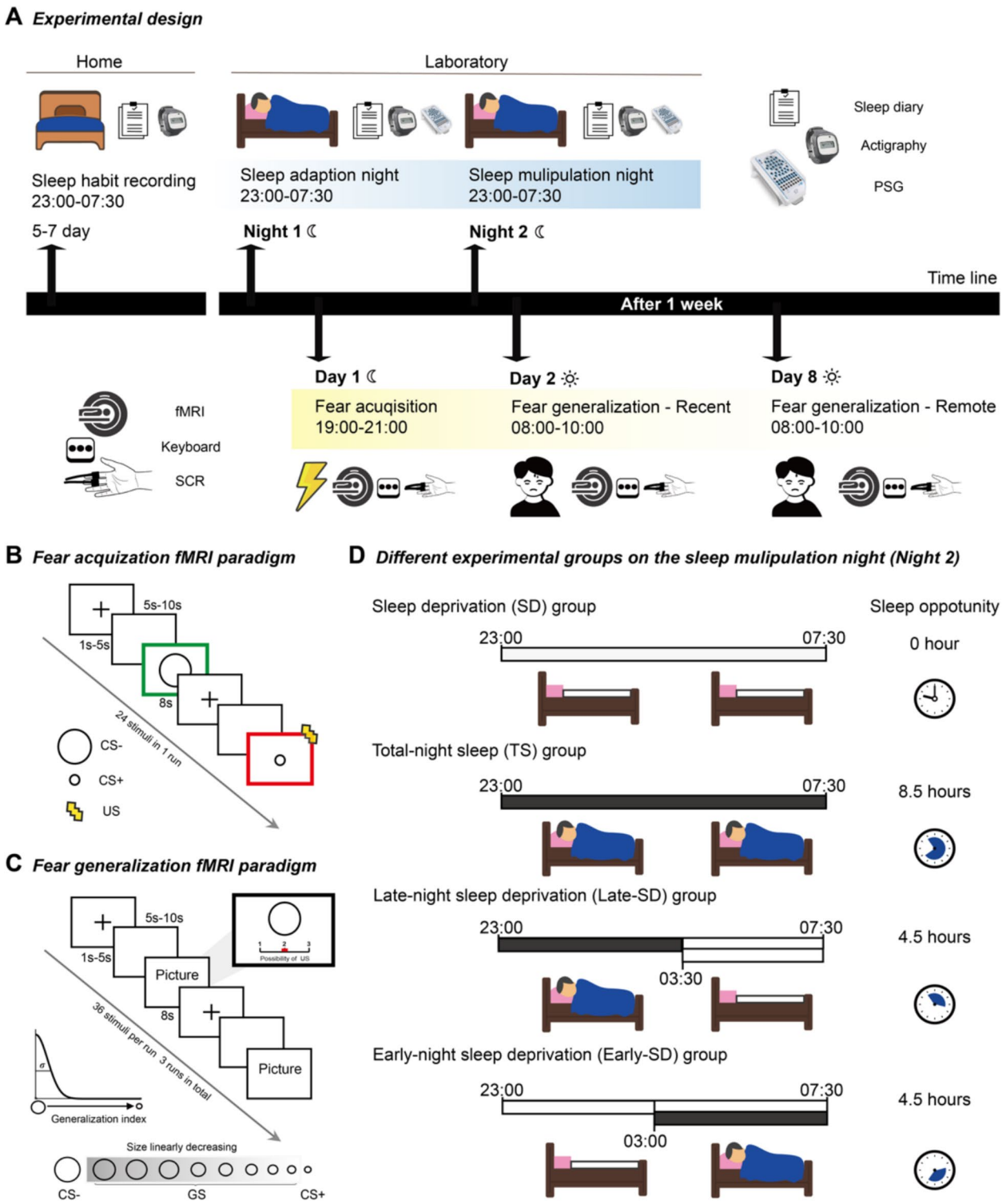

Fig. 1 (See legend on previous page.)

steps [35]. The generalization phase was divided into three runs. In each run, participants were presented with all the stimuli (CS+, CS−, GS), with each stimulus shown 12 times (lasting 8 s each), with 10- to 16-s blank intervals. Participants were instructed to quickly assess the likelihood of receiving an electric shock (US) for each

stimulus, rating it on a 3-point Likert scale ranging from 1 (slightest likelihood) to 3 (strong likelihood), using designated response keys. Participants who showed a uniform response across trails or missed more than 20% of the stimuli were excluded from the analysis. The presentation order was pseudorandomized to ensure that no more than two consecutive stimuli of the same type were shown. The CS+ was followed by an electrical shock twice in each run to minimize extinction effects. All stimulus presentations were controlled using the E-Prime software (Psychology Software Tools, Sharpsburg, PA, USA).

For quantifying the continuous gradient of fear responses across a stimulus continuum, we used the Gaussian curve fit to model how fear responses decline as stimuli become increasingly dissimilar to the conditioned threat cue [36, 37]:

$$f(x) = a^* \exp\left(-(\mathrm{b}-\mathrm{x})^2/2c^2\right)$$

where "*a*" represents the curve's height, "*c*" signifies its width, and "*b*" indicates the peak's location). The parameter "*c*" (denoted as σ) is a key index of the extent of fear generalization (abbreviated as σ), which is the fear generalization index in our analysis (Fig. 1C). Larger σ values indicate higher generalization, whereas smaller σ reflects lower generalization.

Skin conductance response recording and analysis

Skin conductance response (SCR) electrodes were affixed to the second and third fingers of the left hand of each participant using a Biopac MP160 acquisition system (BIOPAC Systems, Inc., Goleta, CA). Conductance was recorded with a gain of 5 μS/V and a 1-Hz low-pass filter during data acquisition. Data collection and analysis were performed using the AcqKnowledge software package (BIOPAC Systems, Inc., Goleta, CA). The analysis focused on an 8-s time window of stimulus-related SCR, defined as the interval starting 2 s after stimulus onset [38]. For each stimulus, the peak-to-peak difference in skin conductance values within this time window was calculated. To ensure consistency with the standard SCR analysis protocols, square-root transformations were applied to the data [39].

Support vector machine analysis

We applied a support vector machine (SVM) with a linear kernel to classify participants based on their fear generalization patterns. Thirteen standardized sleep-related features served as input variables for the analysis. Classification labels were assigned as follows: participants with a fear generalization score exceeding the mean of the SD group were labeled as 1 (high-level overgeneralization), while those with scores below this threshold were labeled as −1 (low-level overgeneralization). To optimize the SVM model, we employed a fivefold cross-validation strategy. This approach ensured robust model performance by iteratively dividing the dataset into five subsets for training and validation to identify the optimal hyperparameters. After model training, we conducted feature weight analysis to evaluate the contribution of each sleep-related feature to the classification. All analyses were performed using Matlab's LIBSVM library (version 3.35. http://www.csie.ntu.edu.tw/~cjlin/libsvm).

Polysomnographic recording and band power analysis

Sleep was recorded using the Somte Polysomnographic (PSG) mobile recording system (Grael, Compumedics Inc., Charlotte, NC), with a montage consisting of six electroencephalography (EEG) channels positioned at F3, F4, C3, C4, O1, and O2, referenced to the contralateral mastoid (A1, A2) in accordance with the International 10–20 system. Bilateral electromyography (EOG) and electrocardiography (EMG) data were concurrently recorded.

Sleep stages were manually scored in 30-s epochs following the American Academy of Sleep Medicine (AASM) standard criteria by two independent sleep technicians blinded to group assignments. Relative band power spectra were analyzed using the well-established open-source sleep algorithm YASA [40]. Sleep stage percentages were prioritized to account for sleep parameter variability induced by the split-night design. Power spectral density (PSD) for each 30-s epoch was calculated using Welch's averaged modified periodogram, incorporating linear detrending, 50% overlap, and Hamming windowing. The PSD was averaged across the following frequency bands: delta (0.5–4.0 Hz), theta (4.0–8.0 Hz), alpha (8.0–12.0 Hz), sigma (12.0–16.0 Hz), and beta (16.0–30.0 Hz). Frontal, parietal, and occipital activities were defined as the averages of F4 and F3, C3 and C4, and O1 and O2 channels, respectively [22].

fMRI acquisition and preprocessing

fMRI data were collected using a 3.0-T GE Discovery MR750 system (General Electric Medical System, Milwaukee, WI, USA) at Peking University Sixth Hospital. Imaging parameters for the echo-planar imaging (EPI) sequence included 33 axial slices (4.2 mm thickness, no gap), repetition time (TR) of 2.00 s, echo time (TE) of 30 ms, flip angle of 90°, in-plane resolution of 3.5×3.5 mm^2, and a field of view (FOV) of 224×224×64 mm^3. High-resolution T1-weighted images were acquired for structural reference, with a voxel size of 1×1×1 mm^3, consisting of 192 slices. Stimuli were presented via back-projection and viewed through a mirror mounted on the head coil.

Data preprocessing was conducted using Statistical Parametric Mapping (SPM12 v7487, https://www.fil.ion.ucl.ac.uk/spm/software/spm12/). The first ten volumes of each run were discarded to allow for T1 equilibration. All functional volumes were realigned to the mean image and co-registered to anatomical images with affine transformations for each participant separately. Two subjects associated with more than 0.3 mm of mean framewise displacement were excluded to minimize the influence of motion artifacts. Anatomical scans were DARTEL-normalized to MNI-space following segmentation. The same transformation was applied to co-registered functional volumes using a small smoothing kernel with 6×6×6 mm of FWHM. Anatomical locations were determined based on the Anatomical Automatic Labeling (AAL) Atlas.

fMRI analysis

fMRI data analysis was performed using the SPM12 toolbox. Two separate subject-level general linear model (GLM) analyses were conducted. The first GLM model aimed to generate beta images for prediction analysis. This model incorporated three distinct regressors time-locked to the presentations of stimuli (i.e., CS+, all GS, CS−), enabling the modeling of brain activity in response to each condition independently. To account for motor activity effects, the model included a boxcar regressor indicating the rating period, with the fixation-cross epoch serving as the implicit baseline. The second model focused on the difference between regressors for the GS and CS− conditions, where GS represented the fear generalization period and CS− served as the baseline. Age and gender were included as parametric modulators in the design matrix to account for their potential influence. All task-related regressors were convolved with the canonical hemodynamic response function. A high-pass filter of 128 s was applied to remove low-frequency drifts. Data from multiple runs were concatenated using SPM's spm_fmri_concatenate.m function, which adjusted for high-pass filter and temporal non-sphericity calculations. Nuisance regressors included 24 motion-related parameters: six head movement parameters, their squares, derivatives, and squared derivatives. Inter-group comparisons at the whole-brain level began with an *F*-test to identify overall differences among the three sleep manipulation groups. Post hoc tests were then conducted to specify which group pairs accounted for significant effects. Statistical significance was set at $p < 0.001$ at the voxel level, with results corrected for family-wise error (FWE) at the cluster level.

The CONN toolbox was used to analyze functional connectivity (FC). Preprocessed data, including motion-corrected, normalized, and denoised time series, were used as input for FC analysis. Regional averaged time series were extracted from the specific regions-of-interest (ROI), and Pearson correlation coefficients were computed to represent FC. Fisher's *r*-to-*z* transformation was applied before statistical analysis. Whole-brain FC patterns were evaluated using the Automated Anatomical Labeling-90 template, to evaluate connectivity patterns across the entire brain, extracting time series data from the seed region and correlating it with time series data from all other voxels in the brain.

Statistical analysis

Statistical analyses were performed using SPSS 26.0 software (SPSS, Chicago, IL, USA). Age and gender were controlled as covariates in all analyses. Repeated-measures ANOVA was used to analyze both between- and within-subject effects for SCR and self-reported fear risking score, followed by the Bonferroni post hoc *t*-tests. To address multiple comparisons within the same stage of analysis, Bonferroni correction was employed to control the family-wise error rate. Linear regression models were used to examine the predictive effects of specific sleep stage percentages on fear generalization, evaluating the explained variance and the significance of individual predictors. Mediation analysis was conducted using the SPSS PROCESS 3.4 toolbox, employing a regression-based approach. Direct, indirect, and total effects were calculated using the product-of-coefficients strategy, and significance was assessed through bootstrapping with 5000 resamples. An indirect effect was considered significant if the 95% bias-corrected confidence intervals from the 5000 bootstrap samples did not cross zero. All results were considered statistically significant at $p < 0.05$ (two-tailed).

Results

Half-night sleep deprivation does not affect the expression of fear memory

To dissociate the functional contributions of REM and NREM sleep to fear generalization processes, we implemented a split-night protocol capitalizing on the circadian distribution of sleep phases. Since the REM sleep predominantly occurs during the second half of the night, the early-night sleep deprivation group was enriched in NREM sleep, while the late-night sleep deprivation group was enriched in REM sleep. A total of 126 participants were included in the final analysis (see experimental protocol in Fig. 1) and assigned to one of four groups: (1) the TS group, (2) the Early-SD group, (3) the Late-SD group, and (4) the SD group. No significant differences were observed among the groups in age, gender, BMI, or baseline sleep characteristics (Table 1 and Additional file 1: Table S1).

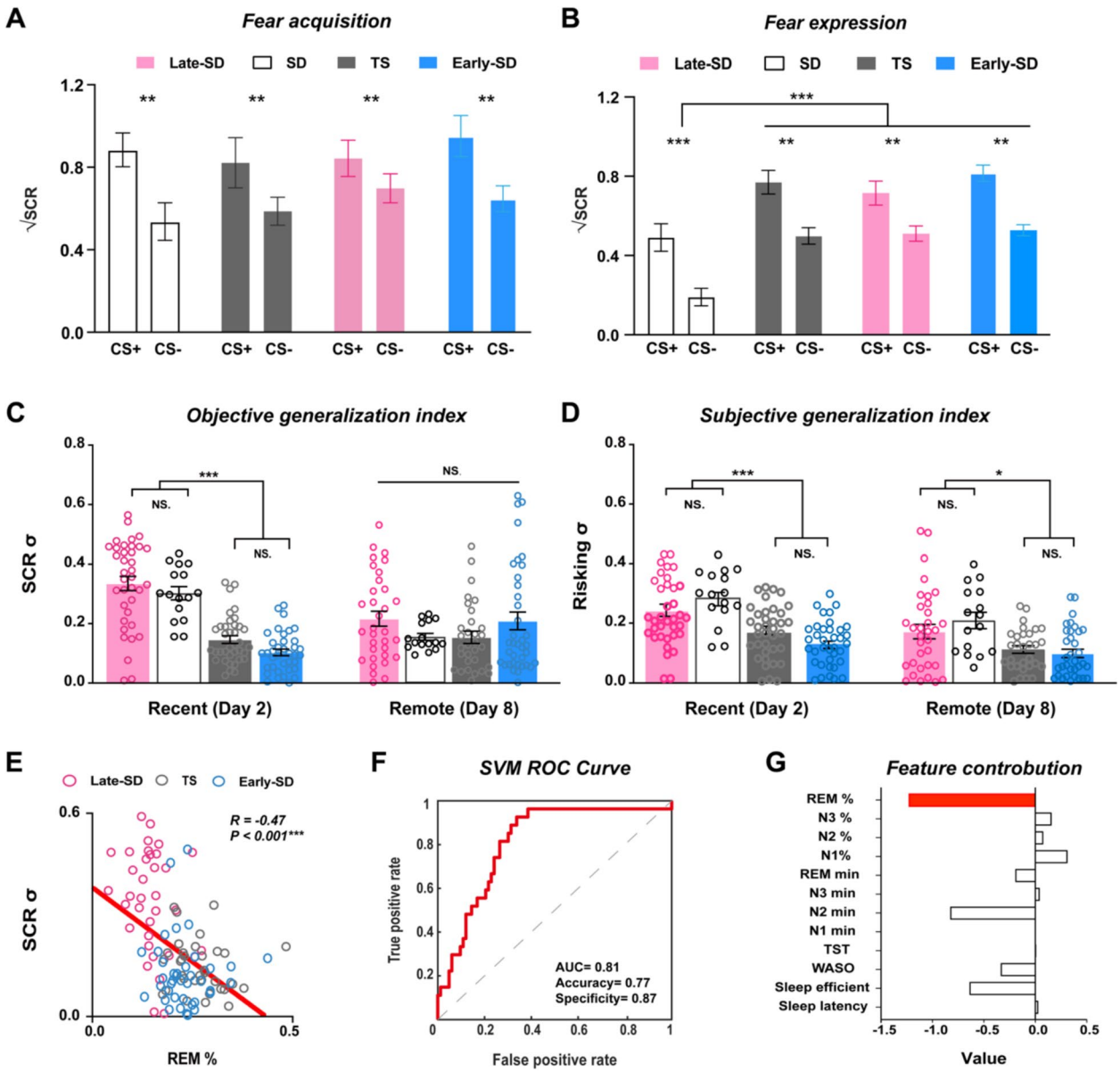

Fig. 2 The impact of sleep manipulation on fear generalization and the mitigating role of REM sleep in overgeneralization. **A** The acquisition of conditioned fear memory before sleep. **B** The expression of fear memory after sleep. **C** and **D** Inter-group differences in objective and subjective fear generalization following sleep and 1 week later. The fear generalization index sigma (σ), derived from objective skin conductance levels (**C**) and subjective risk ratings (**D**), respectively, characterizes the extent of fear generalization. **E** Linear associations between the percentages of REM sleep and the degree of recent fear generalization. **F** The receiver operating characteristic curve for classification of higher and lower overgeneralization using a support vector machine (SVM) model. **G** Extract feature weights from the weight vector of the linear SVM. Bars and error bars represent mean and SEM values, respectively. NS, non-significant. $^{**}p < 0.01$, $^{***}p < 0.001$. Late-SD, late-night sleep deprivation group; Early-SD, early-night sleep deprivation group; TS, total sleep group; SD, total sleep deprivation group; σ, fear generalization gradient index; SCR, skin conductance response; N2%, percentage of N2 sleep; N3%, percentage of N3 sleep; REM %, percentage of REM sleep

Prior to investigating sleep's impact on fear generalization, we first verified successful fear acquisition equivalence across groups. We found that all groups exhibited higher levels of SCR responses to CS+ than CS− during fear acquisition (main effect of stimulus type: $F_{1,122} = 87.55$, $p = 7.50e-06$, Fig. 2A). Specifically, TS group showed $t = 19.61$, $p = 1.13e-05$; Early-SD group showed $t = 19.28$, $p = 4.11e-05$; Late-SD group showed $t = 18.94$, $p = 8.15e-05$. Further, the interaction between stimuli types (CS+, CS−) and groups was not significant

($F_{3,122}=1.40$, $p=0.46$). The above results hence indicated the successful acquisition of conditioned fear without group differences due to sleep manipulation.

Following the sleep manipulation night, no significant differences in fear expression (as assessed via elevated SCR responses) were observed among the three sleep groups (TS, Early-SD, and Late-SD). The interaction between stimuli types (CS+, CS−) and groups was not significant ($F_{3,122}=0.38$, $p=0.76$). However, all three sleep groups exhibited higher levels of SCR compared to the SD group (TS group showed $t=20.22$, $p=3.15e-03$; Early-SD group showed $t=35.96$, $p=8.67e-03$; Late-SD group showed $t=15.68$, $p=4.11e-03$, Fig. 2B). Furthermore, the mean differences in fear expression between CS+ and CS− were not significantly different among the four groups (Additional file 1: Fig. S1), indicating that half-night sleep loss did not affect fear expression.

Sleep deprivation induces fear overgeneralization that could be reversed by REM sleep

We examined the effects of sleep manipulation on both recent (day 2) and remote (day 8) fear generalization. The generalization gradient index (σ) was calculated as the response slopes for all stimuli varying in size from CS+ to CS−, based on both objective SCR and subjective fear ratings (Risking) (see Methods for details).

In the test for recent fear generalization (day 2), the SCR σ showed pronounced group differences (Fig. 2C left, $F_{3,122}=34.04$, $p=4.67e-16$), with Early-SD and TS groups exhibited significantly lower SCR σ compared to Late-SD and SD groups. However, no significant differences were found between the Early-SD and TS groups ($p=0.16$), or between the Late-SD and SD groups ($p=0.68$). This pattern was replicated in subjective responses (Risking σ) (Fig. 2D left, $F_{3,\ 122}=11.41$, $p=1.21e-06$), where Early-SD and TS groups reported significantly lower σ compared to Late-SD and SD groups, with no significant differences between the Early-SD and TS groups ($p=0.19$) or between the Late-SD and SD groups ($p=0.11$). Crucially, the REM-deprived Late-SD group exhibited generalization magnitudes comparable to total sleep deprivation.

In the test for remote fear generalization (day 8), SCR σ group differences dissipated (Fig. 2C right, $F_{3,113}=2.02$, $p=0.12$), whereas Risking σ remained elevated in REM-restricted groups (Fig. 2D right, $F_{3,122}=4.68$, $p=2.72e-02$). The temporal divergence between objective and subjective measures was quantified through differential correlations: highly positive in the test for recent fear generalization (Additional file 1: Fig. S2, $r=0.83$, $p=2.59e-33$) versus null association remotely (Additional file 1: Fig. S2, $r=-0.01$, $p=0.45$). The differential correlation patterns for subjective and objective fear traces may reflect potential dissociations in the consolidation processes of implicit and explicit fear generalization at the remote test.

Sleep architecture analysis confirmed successful REM/NREM sleep-phase manipulation. On the sleep manipulation night, significant differences were observed in the percentages of N2 (Additional file 1: Table S2, $F_{2,107}=7.22$, $p=6.12e-01$), N3 ($F_{2,107}=10.17$, $p=9.07e-05$), and REM sleep stages ($F_{2,107}=32.35$, $p=1.03e-11$) among the three sleep groups (Early-SD, Late-SD, and TS): (1) the Early-SD group exhibited a lower percentage of N2 sleep compared to both the Late-SD ($t=-3.49$, $p=2.52e-03$) and TS ($t=-2.94$, $p=1.16e-02$) groups; (2) both the Late-SD ($t=4.19$, $p=2.35e-04$) and Early-SD ($t=3.49$, $p=2.43e-03$) groups showed a higher percentage of N3 sleep than the TS group; (3) the REM sleep percentage in the Late-SD group ($t=-5.80$, $p=3.82e-07$) was lower than those in the other two groups. These findings (detailed in Additional file 1: Table S2) confirm that the split-night design successfully manipulated REM and NREM sleep proportions. Notably, the Early-SD group displayed a higher percentage of REM sleep than the Late-SD group.

We further explored the predictive effects of specific sleep phase on recent fear generalization. Linear regression identified REM percentage as the principal predictor of recent SCR σ ($\beta=-0.47$, $p=4.79e-07$, $R^2=0.21$) (Fig. 2E). In contrast, no significant predictive effects were found for N2 ($\beta=0.13$, $p=0.19$, $R^2=0.020$) or N3 ($\beta=0.23$, $p=0.051$, $R^2=0.050$) (Additional file 1: Fig. S3). To ensure that the predictive effect of REM percentage on SCR σ was not confounded by group differences, we included group as a covariate in the regression model, and the significant prediction almost remained the same ($\beta=-0.45$, $p=3.08e-01$, $R^2=0.21$), independent of the group effect ($\beta=0.04$, $p=0.62$, $R^2=0.21$). A similar negative predictive effect was observed between REM percentage and subjective risking σ ($\beta=-0.36$, $p=6.99e-03$, $R^2=0.065$) (Additional file 1: Fig. S4). In contrast, neither N2 ($\beta=0.18$, $p=0.07$, $R^2=0.03$) or N3 ($\beta=0.01$, $p=0.91$, $R^2=1.19e-04$) showed statistically or meaningful effect size. These results specifically implicate REM sleep modulation in recent fear generalization processes. So, we prioritized SCR as the main indicator for detecting the immediate neurobiological effects of sleep, while taking the subjective report as a supplementary indicator.

To further validate the specific role of REM phase in fear generalization, we employed a support vector machine (SVM) model for a data-driven analysis of REM sleep's impact (Fig. 2F, AUC = 0.81). Participants from the three sleep groups (Early-SD, Late-SD, and TS) were classified as higher (above) and lower (below)

overgeneralization based on the mean level of fear generalization observed in the SD group. The model incorporated all sleep-related parameters, including the durations and percentages of each sleep stage. Results identified REM percentage as the dominant discriminant feature (Fig. 2G, weight = −1.23) between higher and lower generalization subtypes. Hence, the above findings further confirmed the critical role of the REM phase in modulating fear generalization during sleep.

REM sleep modulates dorsolateral prefrontal cortex activity in fear generalization procession

We next investigated the predictive effects of REM sleep percentage on brain activity during the recent fear generalization test. Among the three sleep groups (Early-SD, Late-SD, and TS), we found significant group differences in brain activation (represented by BOLD signals) in the contrast of generalization stimuli (GS) vs. CS− ($F_{2,107}=5.13$, $p=2.23e-02$, Fig. 3A left). Specifically, the Early-SD group exhibited heightened activation across five clusters, including the right DLPFC (MNI: $x=39$, $y=44$, $z=6$), bilateral middle occipital gyri, lingual gyrus, and cingulate cortex (cluster level $P_{FWE}<0.05$, Fig. 3A right, C and Additional file 1: Table S3). Among these brain clusters, only the right DLPFC activation demonstrated a positive linear relationship with REM percentage across participants ($r=0.43$, $p=2.05e-06$, Fig. 3B). Furthermore, the right DLPFC activation also showed a negative linear correlation with SCR σ ($r=-0.21$, $p=4.73e-03$, Additional file 1: Fig. S5), whereas its association with subjective risk σ was non-significant ($r=-0.04$, $p=0.218$, Additional file 1: Table S4). There were no significant whole-brain BOLD activation differences among the three sleep groups on the remote fear generalization test. Collectively, these findings suggest that reduced cognitive control effect of DLPFC may underlie the association between reduced REM sleep and increased fear generalization following the sleep manipulation.

We further investigated functional connectivity (FC) to determine whether the REM sleep-related effects on fear generalization occurred at both local and large-scale

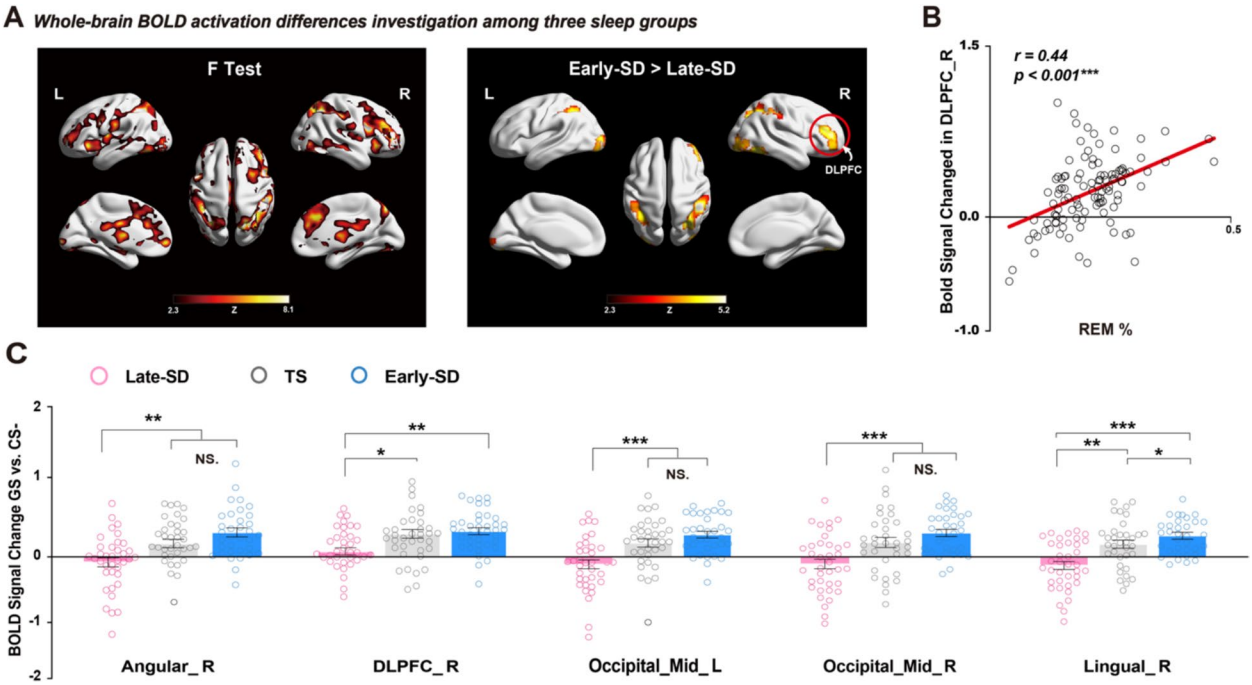

Fig. 3 Differential BOLD activity of fear generalization among the manipulated sleep groups. **A** The left panel highlights brain regions showing significant differences in task-modulated BOLD responses among the three sleep groups at the whole-brain level. On the right, post hoc analysis revealed that the Early-SD group showed significantly higher activity in response to the GS compared to the CS− in the DLPFC_R, Angular_R, Occipital_Mid_L, Occipital_Mid_R, and Lingual_R regions than the Late-SD. FWE corrected $p<0.05$, with the color scale indicating vertex-wise p values. **B** The correlation between the percentage of REM sleep and the BOLD change in the DLPFC_R. **C** Group differences in BOLD activation in the five identified brain regions under GS conditions compared to CS− conditions. Bars and error bars represent mean and SEM values, respectively. BOLD, blood-oxygen-level-dependent; DLPFC_R, right dorsolateral prefrontal cortex; Angular_R, right angular; Occipital_Mid_L, left middle occipital gyrus; Occipital_Mid_R, right middle occipital gyrus; Lingual_R, right lingual; CS−, safe conditioned stimulus; GS, generalization stimulus. *$p<0.05$, **$p<0.01$, ***$p<0.001$. Late-SD, late-night sleep deprivation group; Early-SD, early-night sleep deprivation group; TS, total sleep group; SD, total sleep deprivation group; REM %, percentage of REM sleep

brain network levels. Using the right DLPFC as the seed region, a whole-brain FC analysis revealed no significant group differences (Additional file 1: Fig. S6). We then performed a region-of-interest (ROI-to-ROI) analysis focusing on the five identified generalization-related clusters. This analysis showed that TS participants exhibited stronger connectivity between middle occipital gyrus and DLPFC, angular gyrus, and lingual gyrus compared to both deprivation groups. Notably, Early-SD and Late-SD groups showed comparable connectivity reductions. These findings suggest that occipital-centered fear-generalization FCs are particularly sensitive to sleep restriction rather than phase-specific deprivation.

Frontal theta and delta oscillations exhibit opposing roles in modulating the influence of REM sleep on fear generalization

Building on the critical role of REM phase in fear generalization and the on-line generalization-related DLPFC activity, we further explored the off-line neural mechanisms during sleep. Band power analysis was conducted to examine frequency spectra (delta, theta, alpha, sigma, beta) using PSG recordings. Representative hypnograms and spectrograms for each group are shown in Additional file 1: Fig. S7. Significant differences were observed among the three sleep groups (TS, Early-SD, Late-SD) in delta ($F_{2,106}=4.60$, $p=1.21e-02$) and theta ($F_{2,106}=4.77$, $p=1.04e-02$) power during the REM phase (Fig. 4A). Moreover, control analysis during N3 sleep revealed no significant band power differences among the groups (Additional file 1: Fig. S8), underscoring REM specificity in fear generalization.

We next analyzed the correlations between REM delta and theta band activity and fear generalization. REM theta power demonstrated a robust negative correlation with SCR σ ($r_{channel}=[-0.29,-0.82]$), whereas REM delta power showed paradoxical positive associations ($r_{channel}=[0.24, 0.49]$). These associations were most pronounced in the prefrontal region (average r of F3 and F4 = 0.45, $p=1.59e-22$ for theta power; r of average F3 and F4 = −0.30, $p=3.19e-07$ for delta power; Fig. 4B).

To explore the mechanistic role of these oscillations, a mediation analysis revealed significant mediated effects of both theta and delta power in frontal channels (average F3 and F4) on the correlation between REM percentage and SCR σ. Frontal theta power mediated 44% of REM's protective influence on fear generalization (indirect effect = −0.40, $p=1.24e-07$), while the indirect effect for delta power was −0.17 ($p=2.23e-0.3$), mediating 17% of the total effect (Fig. 4C and D).

Additionally, we examined the potential for a chain mediation model linking REM percentage, frontal theta power, DLPFC activation, and fear generalization. However, this hypothesis was not supported by our data (Additional file 1: Fig. S9), suggesting sleeping-state oscillatory mechanisms operate independently of waking-state BOLD patterns. In summary, these findings indicate that REM theta and delta oscillations have opposing roles in mediating the relationship between REM sleep and fear generalization.

Discussion

To our knowledge, this experiment is the first to examine the specific effects of distinct sleep phases and neural oscillations on both objective and subjective fear generalization in the short (24 h) and long term (1 week), as well as the associated brain activity. This experiment provided new evidence that deprivation of late-night sleep, predominantly REM sleep, results in fear overgeneralization comparable to that observed with whole-night sleep deprivation. Furthermore, the percentage of REM sleep was negatively correlated with fear generalization and positively correlated with generalization-related activity in the DLPFC. Additionally, analysis of neural activity during sleep revealed that theta and delta oscillations during REM sleep acted as key mediators, exerting opposing influences on the relationship between REM sleep and fear generalization. Taken together, these findings provide insight into the critical role of REM sleep in modulating fear generalization and highlight the involvement of the prefrontal cortex in both the sleeping (offline) processing and wakefulness (online) expression of fear generalization.

REM sleep has been identified as a key contributor to attenuating negative emotional arousal [41] and plays a vital role in emotion regulation [42]. Compared to other sleep phases, studies in both animals and humans have shown an increase in REM sleep following learning [43, 44]. Previous research has demonstrated that REM-rich sleep (predominantly occurring in the second half of the night) facilitates successful discrimination between fear-relevant and neutral stimuli [45]. Conversely, selective deprivation of REM sleep has been linked to impairments in memory consolidation [46] and extinction learning [47]. Our findings reveal a distinct and negative correlation between the percentage of REM sleep and fear generalization, with no significant effect of REM sleep percentage on fear expression. Since fear generalization engages higher-order cognitive functions [1], disruptions in REM sleep, commonly observed in individuals with PTSD, are often accompanied by deficits in cognitive flexibility [48]. The reduction in REM sleep may lead to decreased cognitive flexibility and impaired emotional regulation [49]. These findings suggest that preserving REM sleep after a traumatic event, as opposed to NREM

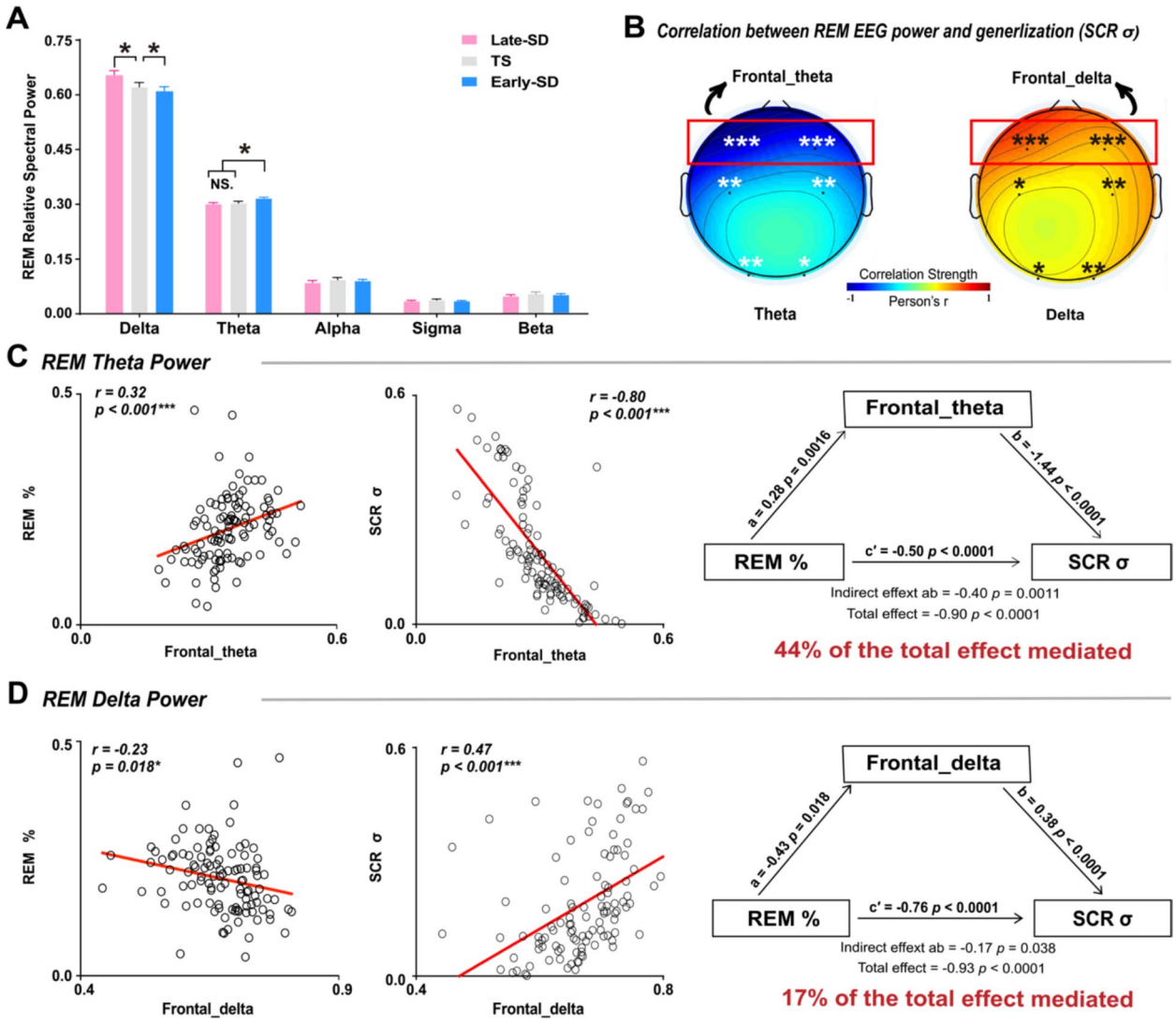

Fig. 4 Brain electrophysiological mechanism underlying the effects of REM sleep on fear generalization. **A** The REM relative spectral power, illustrating delta, theta, alpha, sigma, and beta waves in order of their frequency magnitude. **B** The correlation between relative power and the degree of fear generalization at various electrode sites during REM sleep (left: theta waves, right: delta wave). **C** The correlation and mediation model of theta power in the frontal lobe (F3, F4) for REM % and SCR σ. **D** The correlation and mediation model of delta power in the frontal lobe (F3, F4) for REM % and SCR σ. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. Late-SD, late-night sleep deprivation group; Early-SD, early-night sleep deprivation group; TS, total sleep group; SD, total sleep deprivation group; SCR, skin conductance response; σ, fear generalization gradient index; REM %, percentage of REM sleep

sleep, could be particularly effective in mitigating maladaptive fear generalization and related disorders.

Our study also suggests that REM sleep percentage and duration exert distinct effects on fear generalization. Notably, while the absolute REM duration differed between the TS and Early-SD groups (with the Early-SD group exhibiting a relatively higher REM percentage), both groups showed comparable reductions in fear generalization. This indicates that an elevated REM percentage may exert a protective effect on emotional memory reprocessing, even when absolute REM duration is reduced. Furthermore, our SVM analysis revealed that REM percentage is the dominant feature distinguishing higher from lower generalization subtypes, whereas absolute REM duration does not appear to contribute significantly. This finding further validates the prioritized predictive role of REM percentage—rather than absolute REM duration—in fear generalization. This aligns with clinical evidence demonstrating that alterations in REM percentage can predict PTSD symptom severity [6]. Our

findings underscore the need for future studies to systematically investigate whether REM percentage serves as an adaptive biomarker of emotional regulation capacity.

Impairments in DLPFC function may lead to overestimating threats and making biased judgments, increasing the risk of developing fear-related disorders [50, 51]. We observed that prefrontal cortex neural activity plays a crucial role in both the sleep-mediated modulation and expression of fear generalization. This aligns with findings suggesting the prefrontal cortex is a key node in brain networks that integrate complex generalized memory, highlighting the importance of balance rather than excess [52]. Our results also revealed that the REM-rich Early-SD group exhibited enhanced fear generalization responses in both the middle occipital gyrus and the right angular gyrus compared to the NREM-rich Later-SD group. In a recent study, REM sleep predicts suppression-related engagement of the right DLPFC during deliberate memory suppression [53], a process that shares neural and cognitive parallels with the inhibition of fear generalization. These collectively suggest that REM sleep may facilitate adaptive prefrontal regulation across multiple domains of inhibitory control, including both memory suppression and fear-related responses.

Previous studies have shown that the angular gyrus and the occipital cortex are more activated during threat processing [54, 55]. We recently demonstrated that REM sleep affects not just a single brain region but alters connectivity across multiple networks, including the default mode, cingulo-opercular, auditory, subcortical, and visual networks [34]. Relatedly, research has shown that the angular gyrus encodes and integrates various features of the stimuli [56], while the occipital cortex not only responds to basic visual stimuli [57] but also participates in the cingulo-temporo-occipital circuitry that regulates phobic reactivity [58, 59]. Furthermore, the REM-rich group showed heightened brain activation in the right lingual gyrus during the fear generalization test. This supports findings that REM sleep reduces SCR and enhances lingual gyrus activation in response to extinguished stimuli during fear extinction [13].

Previous studies have reported greater activation in the amygdala and hippocampus in response to fear memory [58, 60, 61]. However, we did not observe differences in brain activity in these regions, which are well known for their role in fear memory, during the fear generalization test. This may be because fear generalization cues likely evoke more complex cognitive processes that require the regulation or inhibition of fear responses [37], rather than the rapid sensory processing of stimuli associated with fear response and memory retrieval, which are primarily facilitated by the amygdala and hippocampus [62]. Another possibility is that, unlike studies involving total sleep deprivation [12, 63], all sleep groups in our analysis had at least 4.5 h of sleep, which may not have significantly impacted the activity of these brain regions [64].

At the neural oscillation level, REM sleep is characterized by faster and low-amplitude theta waves [22, 65]. These theta oscillations resemble the electrophysiological activity observed during wakefulness, in contrast to the delta waves typical of NREM sleep, which are associated with memory suppression or forgetting [66]. Theta wave activity may represent large-scale cooperation between subcortical limbic structures and prefrontal regions, supporting memory-related neuroplasticity [67, 68]. Previous studies have shown enhanced theta activity during the acquisition and retrieval of fear memories [29, 69, 70]. Consistent with this, we identified a negative correlation between theta band activity and fear generalization specifically during REM sleep. This suggests that increased theta oscillations during REM sleep are linked to reduced fear overgeneralization and may help individuals better discriminate between potentially dangerous and safe situations [12]. Our study also revealed that frontal brain regions serve as the anatomical substrates for the effects of theta oscillations in mediating the impact of REM sleep on fear generalization. This finding highlights the role of theta rhythm coordination in enhancing communication between the frontal cortex and deeper subcortical brain regions, which is crucial for processing fear generalization during REM sleep. However, despite the total sleep group spending significantly more time in the REM sleep compared to the REM-rich group, no difference in theta wave power during REM sleep was observed. This may explain why no significant disparity in fear generalization was found between the TS and REM-rich groups.

Limitations and future directions should be noted. First while we included both subjective and objective as fear generalization measures, our analytical focus prioritized SCR due to its established sensitivity to automatic threat detection mechanisms. The observed divergence between remote (day 8) and recent (day 2) generalization differences may reflect distinct temporal dynamics of conscious versus non-conscious fear expression. The divergence findings between SCR versus self-reported fear score may reflect distinct aspects of fear generalization (e.g., conscious vs. autonomic responses), which required further investigation. Another potential limitation of the current study is the lack of direct evidence linking REM sleep to the severity of fear generalization-related mental disorders, such as PTSD, social anxiety disorder, and specific phobias [71, 72]. Another limitation is the absence of simultaneous EEG and fMRI data collection. In this study, the experimental paradigm incorporated both sleep EEG and task fMRI but did not permit the modeling of neural responses to GS or CS+ during

the REM sleep. Future studies could employ integrated fMRI-EEG systems throughout full sleep cycles to provide a more comprehensive understanding of the effects of sleep on fear generalization. Additionally, interventions involving external stimuli, such as targeted memory reactivation through sound, could be used to more precisely explore the pattern trace of on-line fear generalization during off-line sleep period.

Conclusions

In summary, the present study suggests that the reduction of REM sleep following fear conditioning increases the risk of fear overgeneralization and alters related prefrontal activity. It enhances our understanding of the pivotal role of theta oscillations in the prefrontal cortex in mitigating fear overgeneralization during REM sleep. Our findings identify specific sleep phase, electrophysiological activities, and brain targets, opening potential avenues for non-invasive neuromodulation in the treatment of fear overgeneralization-related disorders.

Abbreviations

BOLD	Blood oxygen level-dependent
CS	Conditioned stimulus
CS−	Unconditioned stimulus
CS+	Fear conditioned stimulus
DLPFC	Dorsolateral prefrontal cortex
Early-SD	Early night sleep deprivation group
EEG	Electroencephalogram
fMRI	Functional magnetic resonance imaging
FWE	Family-wise error
GS	Generalization stimulus
Late-SD	Late night sleep deprivation group
SD	Total sleep deprivation group
PSG	Polysomnography
PTSD	Post-traumatic stress disorder
REM	Rapid eye movement sleep
NREM	Non-rapid eye movement sleep
SCR	Skin conductance response
TS	Total sleep group

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12916-025-04411-5.

Additional file 1: Figures S1–S9. Tables S1–S4. Fig. S1 Fear conditioning after sleep. Fig. S2 The linear correlation between objective and subjective fear generalization indices at both the recent and remote fear generalization tests. Fig. S3 Objective skin conductance responses of fear generalization and its relationship to sleep stages. Fig. S4 Subjective risk assessment of fear generalization and its relationship to sleep stages. Fig. S5 The relationship between fear generalization and the BOLD change in the DLPFC_R. Fig. S6 Functional connectivity analysis using the DLPFC as the seed region. Fig. S7 Sleep architecture diagrams for a typical subject from the TS, Early-SD, and Late-SD groups. Fig. S8 Relative spectral power during N3 in different groups. Fig. S9 Chain mediation model combining EEG and fMRI. Table S1 Basic sleep characteristics. Table S2 Sleep parameters during the sleep manipulating night. Table S3 Brain activation in the recent fear generalization task. Table S4 Correlations between five brain regions and fear generalization.

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Authors' contributions

T.D., Y.S., and J.S. conceived and designed the study. T.D., W.L., Y.G., E.Z., Z.Y., and L.S. performed the experiments. T.D., J.L., and Y.S. analyzed and interpreted the data. T.D., T.J., L.L., Y.S., and J.S. wrote and revised the manuscript. All authors contributed to the article and approved the final version of the manuscript.

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Data availability

All data required to support the conclusions of this study are provided within the paper and/or the Additional File 1. The raw data that of this study are available from the corresponding author, Yan Sun and Jie Shi, upon reasonable request.

Declarations

Ethics approval and consent to participate

All experimental procedures were approved by the Ethics Committee at Peking University (The Code of Ethics: IRB00001052-23141). Prior to participation, all individuals provided written informed consent and were compensated monetarily for their involvement in the study.

Consent for publication

All authors have agreed to the submission and publication of this research.

Competing interests

The authors declare no competing interests.

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References

1. Dunsmoor JE, Murphy GL. Categories, concepts, and conditioning: how humans generalize fear. Trends Cogn Sci. 2015;19(2):73–7.
2. Ehlers A, Clark DM. A cognitive model of posttraumatic stress disorder. Behav Res Ther. 2000;38(4):319–45.
3. Shin LM, Liberzon I. The neurocircuitry of fear, stress, and anxiety disorders. Neuropsychopharmacology. 2010;35(1):169–91.
4. Maercker A, Cloitre M, Bachem R, Schlumpf YR, Khoury B, Hitchcock C, et al. Complex post-traumatic stress disorder. Lancet. 2022;400(10345):60–72.
5. Koren D, Arnon I, Lavie P, Klein E. Sleep complaints as early predictors of posttraumatic stress disorder: a 1-year prospective study of injured survivors of motor vehicle accidents. Am J Psychiatry. 2002;159(5):855–7.
6. Slavish DC, Briggs M, Fentem A, Messman BA, Contractor AA. Bidirectional associations between daily PTSD symptoms and sleep disturbances: a systematic review. Sleep Med Rev. 2022;63:101623.

7. Goldstein AN, Walker MP. The role of sleep in emotional brain function. Annu Rev Clin Psychol. 2014;10:679–708.
8. Schafer SK, Wirth BE, Staginnus M, Becker N, Michael T, Sopp MR. Sleep's impact on emotional recognition memory: a meta-analysis of whole-night, nap, and REM sleep effects. Sleep Med Rev. 2020;51:101280.
9. Davidson P, Marcusson-Clavertz D. The effect of sleep on intrusive memories in daily life: a systematic review and meta-analysis of trauma film experiments. Sleep. 2023. https://doi.org/10.1093/sleep/zsac280.
10. Feng P, Becker B, Feng T, Zheng Y. Alter spontaneous activity in amygdala and vmPFC during fear consolidation following 24 h sleep deprivation. Neuroimage. 2018;172:461–9.
11. Menz MM, Rihm JS, Salari N, Born J, Kalisch R, Pape HC, et al. The role of sleep and sleep deprivation in consolidating fear memories. Neuroimage. 2013;75:87–96.
12. Menz MM, Rihm JS, Buchel C. REM sleep is causal to successful consolidation of dangerous and safety stimuli and reduces return of fear after extinction. J Neurosci. 2016;36(7):2148–60.
13. Spoormaker VI, Sturm A, Andrade KC, Schroter MS, Goya-Maldonado R, Holsboer F, et al. The neural correlates and temporal sequence of the relationship between shock exposure, disturbed sleep and impaired consolidation of fear extinction. J Psychiatr Res. 2010;44(16):1121–8.
14. Spoormaker VI, Gvozdanovic GA, Samann PG, Czisch M. Ventromedial prefrontal cortex activity and rapid eye movement sleep are associated with subsequent fear expression in human subjects. Exp Brain Res. 2014;232(5):1547–54.
15. Pace-Schott EF, Milad MR, Orr SP, Rauch SL, Stickgold R, Pitman RK. Sleep promotes generalization of extinction of conditioned fear. Sleep. 2009;32(1):19–26.
16. Manassero E, Giordano A, Raimondo E, Cicolin A, Sacchetti B. Sleep deprivation during memory consolidation, but not before memory retrieval, widens threat generalization to new stimuli. Front Neurosci. 2022;16:902925.
17. Davidson P, Carlsson I, Jonsson P, Johansson M. Sleep and the generalization of fear learning. J Sleep Res. 2016;25(1):88–95.
18. Davidson P, Carlsson I, Jonsson P, Johansson M. A more generalized fear response after a daytime nap. Neurobiol Learn Mem. 2018;151:18–27.
19. Lissek S, Bradford DE, Alvarez RP, Burton P, Espensen-Sturges T, Reynolds RC, et al. Neural substrates of classically conditioned fear-generalization in humans: a parametric fMRI study. Soc Cogn Affect Neurosci. 2014;9(8):1134–42.
20. Kaczkurkin AN, Burton PC, Chazin SM, Manbeck AB, Espensen-Sturges T, Cooper SE, et al. Neural substrates of overgeneralized conditioned fear in PTSD. Am J Psychiatry. 2017;174(2):125–34.
21. Webler RD, Berg H, Fhong K, Tuominen L, Holt DJ, Morey RA, et al. The neurobiology of human fear generalization: meta-analysis and working neural model. Neurosci Biobehav Rev. 2021;128:421–36.
22. Nishida M, Pearsall J, Buckner RL, Walker MP. REM sleep, prefrontal theta, and the consolidation of human emotional memory. Cereb Cortex. 2009;19(5):1158–66.
23. Adamantidis AR, Gutierrez Herrera C, Gent TC. Oscillating circuitries in the sleeping brain. Nat Rev Neurosci. 2019;20(12):746–62.
24. Ramirez-Villegas JF, Besserve M, Murayama Y, Evrard HC, Oeltermann A, Logothetis NK. Coupling of hippocampal theta and ripples with pontogeniculooccipital waves. Nature. 2021;589(7840):96–102.
25. Totty MS, Chesney LA, Geist PA, Datta S. Sleep-dependent oscillatory synchronization: a role in fear memory consolidation. Front Neural Circuits. 2017;11:49.
26. Popa D, Duvarci S, Popescu AT, Lena C, Pare D. Coherent amygdalocortical theta promotes fear memory consolidation during paradoxical sleep. Proc Natl Acad Sci U S A. 2010;107(14):6516–9.
27. Girardeau G, Lopes-Dos-Santos V. Brain neural patterns and the memory function of sleep. Science. 2021;374(6567):560–4.
28. Dejean C, Courtin J, Karalis N, Chaudun F, Wurtz H, Bienvenu TC, et al. Prefrontal neuronal assemblies temporally control fear behaviour. Nature. 2016;535(7612):420–4.
29. Boyce R, Glasgow SD, Williams S, Adamantidis A. Causal evidence for the role of REM sleep theta rhythm in contextual memory consolidation. Science. 2016;352(6287):812–6.
30. Walker MP, van der Helm E. Overnight therapy? The role of sleep in emotional brain processing. Psychol Bull. 2009;135(5):731–48.
31. Krause AJ, Simon EB, Mander BA, Greer SM, Saletin JM, Goldstein-Piekarski AN, et al. The sleep-deprived human brain. Nat Rev Neurosci. 2017;18(7):404–18.
32. Larson O, Schapiro AC, Gehrman PR. Effect of sleep manipulations on intrusive memories after exposure to an experimental analogue trauma: a meta-analytic review. Sleep Med Rev. 2023;69:101768.
33. Lonsdorf TB, Klingelhofer-Jens M, Andreatta M, Beckers T, Chalkia A, Gerlicher A, et al. Navigating the garden of forking paths for data exclusions in fear conditioning research. Elife. 2019. https://doi.org/10.7554/eLife.52465.
34. Di T, Zhang L, Meng S, Liu W, Guo Y, Zheng E, et al. The impact of REM sleep loss on human brain connectivity. Transl Psychiatry. 2024;14(1):270.
35. Hennings AC, McClay M, Drew MR, Lewis-Peacock JA, Dunsmoor JE. Neural reinstatement reveals divided organization of fear and extinction memories in the human brain. Curr Biol. 2022;32(2):304–14 e305.
36. Tuominen L, Boeke E, DeCross S, Wolthusen RP, Nasr S, Milad M, et al. The relationship of perceptual discrimination to neural mechanisms of fear generalization. Neuroimage. 2019;188:445–55.
37. Onat S, Buchel C. The neuronal basis of fear generalization in humans. Nat Neurosci. 2015;18(12):1811–8.
38. Lonsdorf TB, Menz MM, Andreatta M, Fullana MA, Golkar A, Haaker J, et al. Don't fear 'fear conditioning': methodological considerations for the design and analysis of studies on human fear acquisition, extinction, and return of fear. Neurosci Biobehav Rev. 2017;77:247–85.
39. Kuhn M, Gerlicher AMV, Lonsdorf TB. Navigating the manyverse of skin conductance response quantification approaches - a direct comparison of trough-to-peak, baseline correction, and model-based approaches in Ledalab and PsPM. Psychophysiology. 2022;59(9):e14058.
40. Vallat R, Walker MP. An open-source, high-performance tool for automated sleep staging. Elife. 2021. https://doi.org/10.7554/eLife.70092.
41. Hutchison IC, Pezzoli S, Tsimpanouli ME, Abdellahi MEA, Pobric G, Hulleman J, et al. Targeted memory reactivation in REM but not SWS selectively reduces arousal responses. Commun Biol. 2021;4(1):404.
42. Schwartz S, Clerget A, Perogamvros L. Enhancing imagery rehearsal therapy for nightmares with targeted memory reactivation. Curr Biol. 2022;32(22):4808–16 e4804.
43. Rasch B, Born J. About sleep's role in memory. Physiol Rev. 2013;93(2):681–766.
44. Li W, Ma L, Yang G, Gan WB. REM sleep selectively prunes and maintains new synapses in development and learning. Nat Neurosci. 2017;20(3):427–37.
45. Diekelmann S, Wilhelm I, Wagner U, Born J. Sleep to implement an intention. Sleep. 2013;36(1):149–53.
46. Kumar D, Koyanagi I, Carrier-Ruiz A, Vergara P, Srinivasan S, Sugaya Y, Kasuya M, Yu TS, Vogt KE, Muratani M, et al. Sparse activity of hippocampal adult-born neurons during REM sleep is necessary for memory consolidation. Neuron. 2020;107(3):552–65 e510.
47. Spoormaker VI, Schroter MS, Andrade KC, Dresler M, Kiem SA, Goya-Maldonado R, et al. Effects of rapid eye movement sleep deprivation on fear extinction recall and prediction error signaling. Hum Brain Mapp. 2012;33(10):2362–76.
48. Zhang Y, Ren R, Sanford LD, Yang L, Zhou J, Zhang J, et al. Sleep in posttraumatic stress disorder: a systematic review and meta-analysis of polysomnographic findings. Sleep Med Rev. 2019;48:101210.
49. Walker MP, Liston C, Hobson JA, Stickgold R. Cognitive flexibility across the sleep-wake cycle: REM-sleep enhancement of anagram problem solving. Brain Res Cogn Brain Res. 2002;14(3):317–24.
50. Roberts AC. Prefrontal regulation of threat-elicited behaviors: a pathway to translation. Annu Rev Psychol. 2020;71:357–87.
51. Wen Z, Seo J, Pace-Schott EF, Milad MR. Abnormal dynamic functional connectivity during fear extinction learning in PTSD and anxiety disorders. Mol Psychiatry. 2022;27(4):2216–24.
52. Cummings KA, Clem RL. Prefrontal somatostatin interneurons encode fear memory. Nat Neurosci. 2020;23(1):61–74.
53. Harrington MO, Karapanagiotidis T, Phillips L, Smallwood J, Anderson MC, Cairney SA. Memory control deficits in the sleep-deprived human brain. Proc Natl Acad Sci U S A. 2025;122(1):e2400743122.

54. Pourtois G, Schwartz S, Seghier ML, Lazeyras F, Vuilleumier P. Neural systems for orienting attention to the location of threat signals: an event-related fMRI study. Neuroimage. 2006;31(2):920–33.
55. Lange I, Goossens L, Bakker J, Michielse S, Marcelis M, Wichers M, et al. Functional neuroimaging of associative learning and generalization in specific phobia. Prog Neuropsychopharmacol Biol Psychiatry. 2019;89:275–85.
56. Ramanan S, Piguet O, Irish M. Rethinking the role of the angular gyrus in remembering the past and imagining the future: the contextual integration model. Neuroscientist. 2018;24(4):342–52.
57. Wudarczyk OA, Kohn N, Bergs R, Goerlich KS, Gur RE, Turetsky B, et al. Chemosensory anxiety cues enhance the perception of fearful faces - an fMRI study. Neuroimage. 2016;143:214–22.
58. Izquierdo I, Furini CR, Myskiw JC. Fear memory. Physiol Rev. 2016;96(2):695–750.
59. Ressler KJ, Berretta S, Bolshakov VY, Rosso IM, Meloni EG, Rauch SL, et al. Post-traumatic stress disorder: clinical and translational neuroscience from cells to circuits. Nat Rev Neurol. 2022;18(5):273–88.
60. Herry C, Johansen JP. Encoding of fear learning and memory in distributed neuronal circuits. Nat Neurosci. 2014;17(12):1644–54.
61. Zhang X, Kim J, Tonegawa S. Amygdala reward neurons form and store fear extinction memory. Neuron. 2020;105(6):1077–93 e1077.
62. Buchel C, Morris J, Dolan RJ, Friston KJ. Brain systems mediating aversive conditioning: an event-related fMRI study. Neuron. 1998;20(5):947–57.
63. Grezes J, Erblang M, Vilarem E, Quiquempoix M, Van Beers P, Guillard M, et al. Impact of total sleep deprivation and related mood changes on approach-avoidance decisions to threat-related facial displays. Sleep. 2021. https://doi.org/10.1093/sleep/zsab186.
64. Hudson AN, Van Dongen HPA, Honn KA. Sleep deprivation, vigilant attention, and brain function: a review. Neuropsychopharmacology. 2020;45(1):21–30.
65. Lee MG, Hassani OK, Alonso A, Jones BE. Cholinergic basal forebrain neurons burst with theta during waking and paradoxical sleep. J Neurosci. 2005;25(17):4365–9.
66. Kim J, Gulati T, Ganguly K. Competing roles of slow oscillations and delta waves in memory consolidation versus forgetting. Cell. 2019;179(2):514–26 e513.
67. Jones MW, Wilson MA. Theta rhythms coordinate hippocampal-prefrontal interactions in a spatial memory task. PLoS Biol. 2005;3(12):e402.
68. Dunn SLS, Town SM, Bizley JK, Bendor D. Behaviourally modulated hippocampal theta oscillations in the ferret persist during both locomotion and immobility. Nat Commun. 2022;13(1):5905.
69. Diekelmann S, Born J. The memory function of sleep. Nat Rev Neurosci. 2010;11(2):114–26.
70. Andrillon T, Nir Y, Cirelli C, Tononi G, Fried I. Single-neuron activity and eye movements during human REM sleep and awake vision. Nat Commun. 2015;6:7884.
71. McTeague LM, Lang PJ. The anxiety spectrum and the reflex physiology of defense: from circumscribed fear to broad distress. Depress Anxiety. 2012;29(4):264–81.
72. McLenon J, Rogers MAM. The fear of needles: a systematic review and meta-analysis. J Adv Nurs. 2019;75(1):30–42.

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